in the hyperplane $\{\boldsymbol{v} \in \mathbb{R}^r : \boldsymbol{c}\boldsymbol{v} = 0\}$.

We get dependences on $V/\boldsymbol{u}_i$ from those on V by simply deleting the components corresponding to $\boldsymbol{u}_i$, while value vectors of $V/\boldsymbol{u}_i$ are those value vectors V which are zero on $\boldsymbol{u}_i$ (i.e., have zero i-component).

Proposition 6.12. *The oriented matroid $\mathcal{M}(V)/i$ of $V/\boldsymbol{u}_i$ is given as follows:*

$$\mathcal{V}(V/\boldsymbol{u}_i) = \{\mathrm{v}\backslash i : \mathrm{v} \in \mathcal{V}(V)\} \qquad \mathcal{V}^*(V/\boldsymbol{u}_i) = \{\mathrm{v}\backslash i : \mathrm{v} \in \mathcal{V}^*(V),\, v_i = 0\}$$

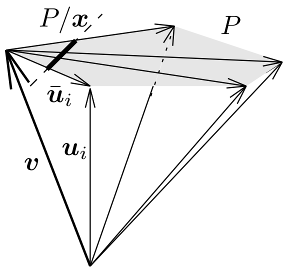
$$\mathcal{C}(V/\boldsymbol{u}_i) = \mathrm{MIN}\{\mathrm{c}\backslash i : \mathrm{c} \in \mathcal{C}(V)\} \quad \mathcal{C}^*(V/\boldsymbol{u}_i) = \{\mathrm{c}\backslash i : \mathrm{c} \in \mathcal{C}^*(V),\, c_i = 0\}.$$

Let us mention two examples that show how deletion and contraction appear in connection with polytopes.

Examples 6.13. Let $P \subseteq \mathbb{R}^d$ be a polytope, let $X := \mathrm{vert}(P)$ be its vertex set, and let $V \in \mathbb{R}^{r \times n}$ be the corresponding vector configuration in $\mathbb{R}^r$ $(r = d+1)$.

If F is a face of P, then the vector configuration for F is obtained by deleting from V all the vertices that do not lie on F.

If $\boldsymbol{x} \in \mathrm{vert}(P) \subseteq \mathbb{R}^d$ is a vertex of P and $\boldsymbol{v} \in V \subseteq \mathbb{R}^r$ is the corresponding vector, then the vector configuration of the vertex figure $P/\boldsymbol{x}$ is the contraction $V/\boldsymbol{v}$. (In this case the projection hyperplane for the contraction can be taken parallel to the hyperplane spanned by P.)



Note that by contracting we get a vector configuration that may also represent a lot of interior points of the vertex figure, corresponding to vertices $\boldsymbol{w} \in \mathrm{vert}(P)$ such that $[\boldsymbol{w}, \boldsymbol{v}]$ is not an edge of P.

6.4 Dual Configurations and Gale Diagrams

Now observe that $\mathrm{Dep}(V) \subseteq \mathbb{R}^n$ determines the configuration $V \in \mathbb{R}^{r \times n}$ of column vectors uniquely up to coordinate transformations in $\mathbb{R}^r$, which correspond to row operations on the matrix V. Thus the dual space $\mathrm{Val}(V)$ determines a configuration of row vectors in $(\mathbb{R}^{n-r})^*$, which completely encodes the vector configuration V.

Theorem 6.14 (Dual configuration).
Let $V \in \mathbb{R}^{r\times n}$ be a configuration of n column vectors in $\mathbb{R}^r$.

Then there is a matrix $G \in \mathbb{R}^{n\times(n-r)}$ of n row vectors in $(\mathbb{R}^{n-r})^*$, such that

$$\mathrm{Val}(V) \;=\; \{\boldsymbol{c} \in (\mathbb{R}^n)^* : \boldsymbol{c}G = \mathbb{0}\}$$

and

$$\mathrm{Dep}(V) \;=\; \{G\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^{n-r}\}.$$

The configuration of row vectors G is uniquely determined by either of the two conditions, up to linear coordinate transformations in $(\mathbb{R}^{n-r})^*$, which correspond to column operations on the matrix G.

Proof. The matrix $G \in \mathbb{R}^{n\times(n-r)}$ has to satisfy

$$\mathrm{rank}(G) = n - r \qquad \textit{and} \qquad VG = O,$$

where O is the zero-matrix in $\mathbb{R}^{r\times(n-r)}$.

For computation, this means that we have to find a basis for the orthogonal complement of the space spanned by the rows of V in $(\mathbb{R}^n)^*$. This is computationally easy: it only requires to get V into a normal form like $V = (I_r|M)$, so that the dual configuration can be obtained as $G := \binom{M}{-I_{n-r}}$.

Existence of G, and uniqueness up to column operations, follows from this. □

If we define the spaces of dependences and of value vectors for configurations of row vectors in exact analogy to the case of column vectors, then we can read Theorem 6.1 as saying that there is a dual configuration G, essentially unique, such that

$$\mathrm{Dep}(V) \;=\; \mathrm{Val}(G)$$

and

$$\mathrm{Val}(V) \;=\; \mathrm{Dep}(G).$$

The following corollary has the additional information that the combinatorics (in particular, the circuits and cocircuits) of the vector configuration V can be read off not only from G, but in fact from the combinatorics of G.

Corollary 6.15. *The circuits of a vector configuration $V \in \mathbb{R}^{r\times n}$ are the cocircuits of the dual configuration $G \in \mathbb{R}^{n\times(n-r)}$, and vice versa.*

In particular, the oriented matroid of the dual configuration is determined by the oriented matroid itself; it is the dual oriented matroid

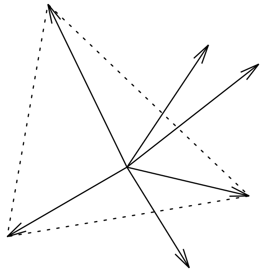
$$\mathcal{M}(G) \;=\; (\mathcal{M}(V))^*.$$

Proof. From Corollary 6.9 and Section 6.3(c). □

In particular, this implies that the dual of an acyclic vector configuration is a *totally cyclic* configuration $G = \{\boldsymbol{g}_1, \ldots, \boldsymbol{g}_n\}$ of row vectors: this property is characterized by the following two properties (which are equivalent by the Farkas lemma):

(i) There is no nonnegative value vector, i.e., no $\boldsymbol{x} \in \mathbb{R}^{n-r}$ with $G\boldsymbol{x} \geq \mathbf{0}$ and $G\boldsymbol{x} \neq \mathbf{0}$.

(ii) There is a positive dependence, i.e., some $\boldsymbol{c} > \mathbb{O}$ with $\boldsymbol{c}G = \mathbb{O}$.



Comparison between these descriptions of "totally cyclic" and the corresponding ones that we have given for the dual concept "acyclic" on page 156 might give a feel for how the translation between dual concepts works on the linear algebra level.

On the combinatorial side, we derive the following.

Corollary 6.16. *A vector configuration V is acyclic if and only if the following equivalent conditions hold:*

(i) *$\mathcal{M}(V)$ has no positive signed circuit.*

(ii) *$(++\ldots+)$ is a signed covector of $\mathcal{M}(V)$.*

(iii) *Every i is contained in a nonnegative cocircuit.*

Dually, a row vector configuration G is totally cyclic if and only if the following equivalent conditions hold:

(i) *$\mathcal{M}(G)$ has no positive signed cocircuit.*

(ii) $\begin{pmatrix} + \\ \vdots \\ + \end{pmatrix}$ *is a signed vector of $\mathcal{M}(G)$.*

(iii) *Every i is contained in a nonnegative circuit.*

In particular, a vector configuration V is acyclic if and only if its dual configuration G is totally cyclic, and conversely.

Now we'll put the pieces together.

Definition 6.17 (Linear and affine Gale diagrams).
Let $P = \operatorname{conv}\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ be a d-polytope in $\mathbb{R}^d$ with n vertices. A *Gale diagram* and an *affine Gale diagram* of P are obtained by the following sequence of operations.

$$\begin{array}{ll}
d\text{-polytope with } n \text{ vertices } \boldsymbol{x}_i \ (1 \le i \le n) & \textit{(polytope)} \\
\Big\updownarrow & \\
n \text{ vectors in } \mathbb{R}^r = \mathbb{R}^{d+1},\ \boldsymbol{v}_i := \binom{1}{\boldsymbol{x}_i} & \\
\Bigg\updownarrow \ * \ \ \text{(oriented matroid duality)} & \\
n \text{ vectors } \boldsymbol{g}_i \text{ in } \mathbb{R}^{n-r} = \mathbb{R}^{n-d-1} & \textit{Gale diagram} \\
\Big\updownarrow & \\
n \text{ (signed) points } \boldsymbol{a}_i \text{ in affine } (n-d-2)\text{-space} & \textit{affine Gale diagram}
\end{array}$$

Here the passage from $\mathbb{R}^d$ to $\mathbb{R}^{d+1}$ is the usual embedding, used to linearize the situation. The dual configuration of this vector configuration is a *Gale diagram* for P, determined uniquely up to a change of coordinates.

For the reduction of $(\mathbb{R}^{n-d-1})^*$ to $(\mathbb{R}^{n-d-2})^*$, we find a suitable vector $\boldsymbol{y} \in \mathbb{R}^{n-d-1}$ such that $\boldsymbol{g}_i\boldsymbol{y} \neq 0$ unless $\boldsymbol{g}_i = \mathbb{0}$, for all i. Then we associate with $\boldsymbol{g}_i$ the point

$$\boldsymbol{a}_i \ := \ \frac{\boldsymbol{g}_i}{\boldsymbol{g}_i\boldsymbol{y}} \ \in \ \{\boldsymbol{c} \in (\mathbb{R}^{n-d-1})^* : \boldsymbol{c}\boldsymbol{y} = 1\} \ \cong \ (\mathbb{R}^{n-d-2})^*,$$

which we call a *positive point* in the affine space $\mathbb{R}^{n-d-2}$ if $\boldsymbol{g}_i\boldsymbol{y} > 0$, and a *negative point* if $\boldsymbol{g}_i\boldsymbol{y} < 0$.

This yields the *affine Gale diagram*, a labeled point configuration

$$\{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\} \subseteq \mathbb{R}^{n-d-2},$$

where the point $\boldsymbol{a}_i$ is labeled i if it is a positive point, labeled $\bar{i}$ if it represents a negative point, and not represented by a point (or, represented by a "special" point) in the case where $\boldsymbol{g}_i = \boldsymbol{0}$.

The reduction to $(\mathbb{R}^{n-d-2})^*$ does not lose combinatorial information: the circuits and cocircuits of this affine point configuration still represent the cocircuits and circuits of P. This is extremely useful for polytopes with "few vertices," where $n - d$ is small, as we will see in the following section. Let us consider one example, to illustrate the technique.

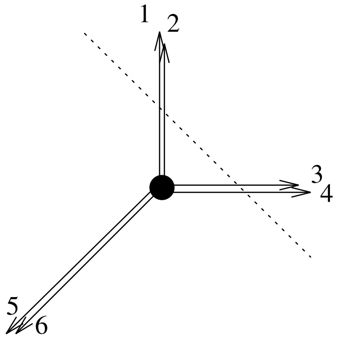
Example 6.18. For the octahedra of Example 6.3, we have the matrices

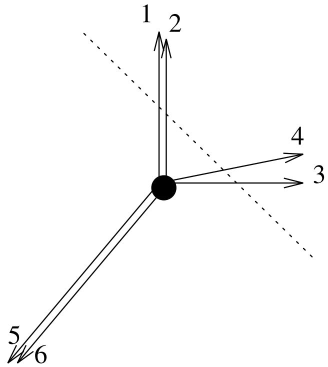
$$V_1 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix} \qquad V_2 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ \frac{1}{6} & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

and compute Gale transforms

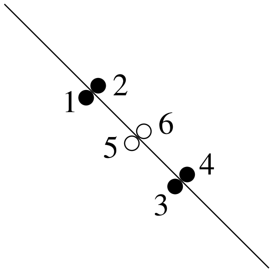
$$G_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ -1 & -1 \\ -1 & -1 \end{pmatrix} \qquad G_2 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \\ 1 & \frac{1}{6} \\ -1 & -\frac{13}{12} \\ -1 & -\frac{13}{12} \end{pmatrix}.$$

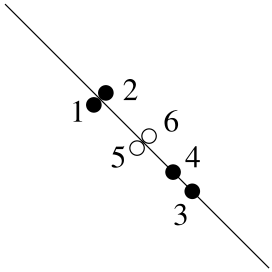
From this we can directly draw Gale diagrams (they are 2-dimensional), and derive 1-dimensional affine Gale diagrams, for $\boldsymbol{y} = \binom{1}{1}$. Here they are, linear and affine.





We use the convention for affine Gale diagrams that black dots denote positive points, while white dots denote negative points.





It is really important that the reader figure out how to read off the circuits and the cocircuits of the octahedra from their affine Gale diagrams: he or she might appreciate that this is a very compact (1-dimensional!) encoding of the combinatorics of 3-dimensional geometric figures.

The basic pattern is as before. Namely, from affine linear functions we read off sign vectors like

$$\begin{pmatrix} 0 \\ 0 \\ + \\ + \\ - \\ - \end{pmatrix}$$

which are cocircuits for both Gale diagrams, and thus circuits of the octahedra. Similarly, from minimal affine dependences we read off sign vectors like

$$(+\,0\,0\,+\,+\,0),$$

which form circuits for the Gale diagram, and thus cocircuits for the octahedra. The only new feature is that the sign is reversed for any negative point in the diagram.

Every spanning set $G = \{\boldsymbol{g}_1, \ldots, \boldsymbol{g}_n\}$ of n row vectors in $(\mathbb{R}^{n-r})^*$ can be interpreted as the Gale diagram of a vector configuration of n vectors that span $\mathbb{R}^d$. However, these vectors need not come from a $(d-1)$-polytope: the vector configuration might not be "affine" (acyclic), and even if it is, the vectors need not be in convex position. However, there is a simple combinatorial condition that characterizes Gale diagrams, see the following theorem. It is important because it allows us to conclude the existence of a (high-dimensional) polytope with specific properties from a (low-dimensional) configuration of signed points.

Theorem 6.19 (Characteristic property of Gale diagrams). *A matrix $G \in \mathbb{R}^{n\times(n-r)}$ of row vectors (of full rank $n-r$) is a Gale diagram of a $(r-1)$-polytope with n vertices if and only if every cocircuit has at least two positive elements.*

Proof. Every spanning configuration G of row vectors is the dual configuration of some spanning vector configuration V in $\mathbb{R}^r$. The configuration V is acyclic if and only if G is totally cyclic, that is, if G has no negative cocircuit: every cocircuit of G has at least one positive element.

With this, we can scale the vectors of V, without changing the combinatorics, such that V comes from a point configuration in some affine hyperplane $H \cong \mathbb{R}^{r-1}$ in $\mathbb{R}^r$. The points in H are in convex position unless one point is in the convex hull of the others, that is, unless V has a circuit with exactly one positive element, and thus G has a cocircuit with exactly one positive element. □

It is easy to translate this condition to affine Gale diagrams. Check it for the affine Gale diagrams of the two octahedra given earlier in Example 6.18!

It yields a criterion that is very easy to check "by inspection" for the interesting case $n - d = 4$; see below.

Corollary 6.20 (Characterization of Gale diagrams of polytopes). *A configuration* $A = \{\boldsymbol{a}_1, \dots, \boldsymbol{a}_n\}$ *points in* $(\mathbb{R}^{n-d-2})^*$*, each of them declared to be either "positive" or "negative," that affinely spans* $(\mathbb{R}^{n-d-2})^*$*, is the Gale diagram of a* d*-polytope with* n *vertices if and only if the following condition is satisfied: for every oriented hyperplane* H *in* $(\mathbb{R}^{n-d-2})^*$ *spanned by points of* A*, the number of positive* A*-points on the positive side of* H*, plus the number of negative* A*-points on the negative side of* H*, is at least* 2.

In our descriptions we have disregarded the case of "special" points: they are just the cone points, so adding k special points to the diagram G corresponds to taking the k-fold pyramid over the polytope represented by G.

6.5 Polytopes with Few Vertices

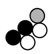
Any d-polytope with $d + 1$ vertices is a d-simplex: this we know and have seen before. In this case the Gale diagram is in 0-dimensional space, so all vectors are $\mathbf{0}$-vectors trivially.

Next consider the case of d-polytopes with $d + 2$ vertices. The result is that there are $\lfloor d^2/4 \rfloor$ combinatorial types of d-polytopes with $d+2$ vertices. Of those, $\lfloor d/2 \rfloor$ represent simplicial polytopes, and the others are (multiple) pyramids over simplicial polytopes of this type. The case of $d + 2$ vertices is classic and can be found in Schoute [480] and Sommerville [506]; see also Grünbaum [252, Sect. 6.1] or Ewald [201, Sect. 2.6].

The affine Gale diagrams representing d-polytopes with $d + 2$ vertices are 0-dimensional and may be represented by a "cloud" of positive points (black), negative points (white), and special points (grey). The condition of Corollary 6.20 requires that there are at least 2 black and at least 2 white points. Furthermore, interchanging black and white points does not change the combinatorial type of the polytope.

Thus we get the following complete enumeration for $d = 3$, $n = 5$:

represents the bipyramid over a triangle (this is a simplicial polytope with 6 facets), and



yields the square pyramid (a nonsimplicial polytope with 5 facets).

Similarly, you should analyze and classify the 4-polytopes with 6 vertices in Exercises 6.8.